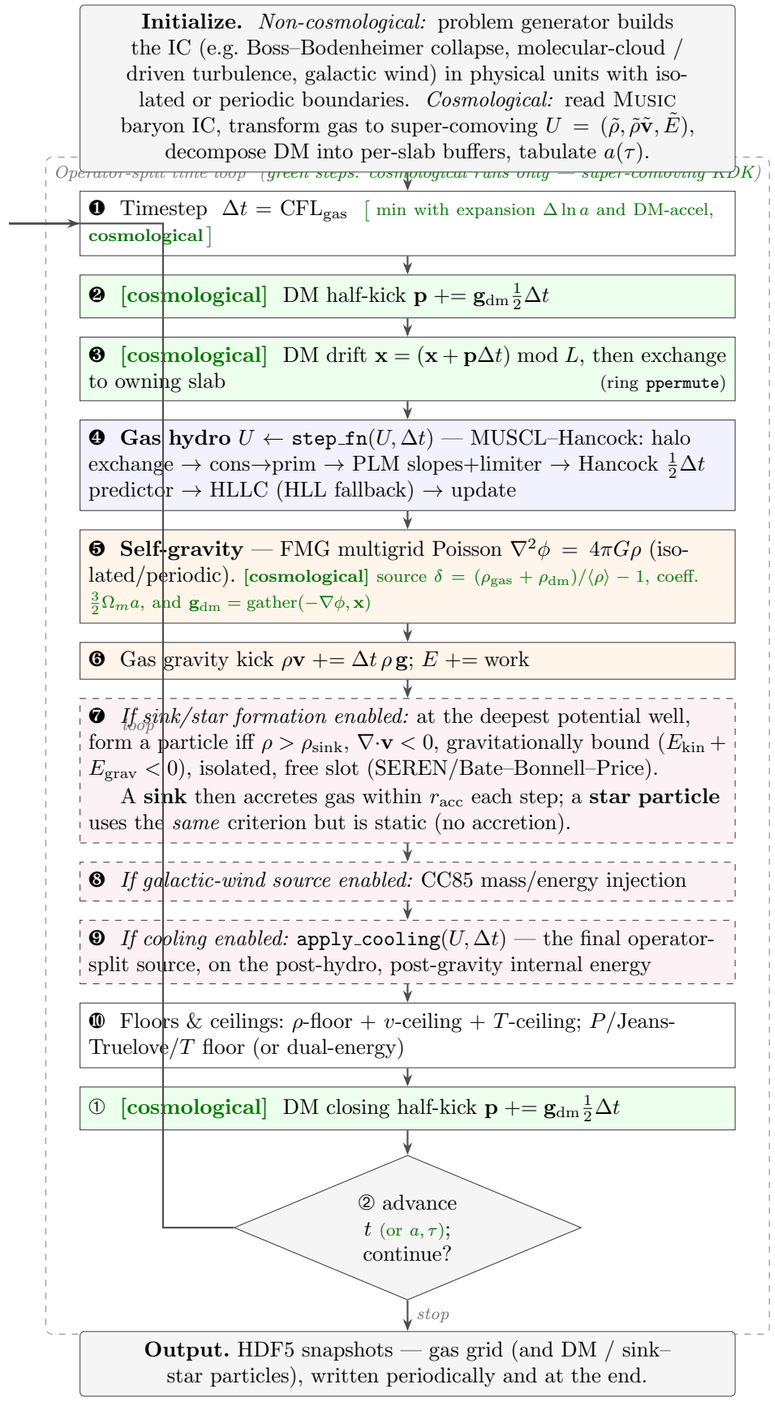


Figure 1: Control flow of LUNA in its most complete configuration. A single operator-split time loop drives both non-cosmological runs (physical time, isolated or periodic boundaries) and cosmological gas+dark-matter runs; the green-tagged steps (DM kick–drift–kick, the expansion and DM-acceleration timestep terms, the super-comoving source and coefficient, and the a, τ advance) are active only in the cosmological mode. The gas hydrodynamics (blue) and self-gravity (orange) are common to both and together form the end-to-end differentiable forward map. The dashed purple modules are optional and applied after the hydrodynamic update and gravitational source, before the floors: sink and star-particle formation share a single criterion—a bound, converging density peak at the deepest potential well—differing only in whether the particle subsequently accretes gas; the CC85 galactic-wind injection and radiative cooling are likewise optional operator-split sources.